A tap crossing a residue carries one whole block

Ordinary coefficient taps

$$b_m = a_m + 2a_{m-1} - a_{m-2}$$

Base 3 phases

$$x_j[n] = a_{3n+j}$$

$$X_j(t) = \sum_n x_j[n] t^n$$

Here t is the formal OGF variable of the block index.

$$Y(t) = H(t)X(t)$$

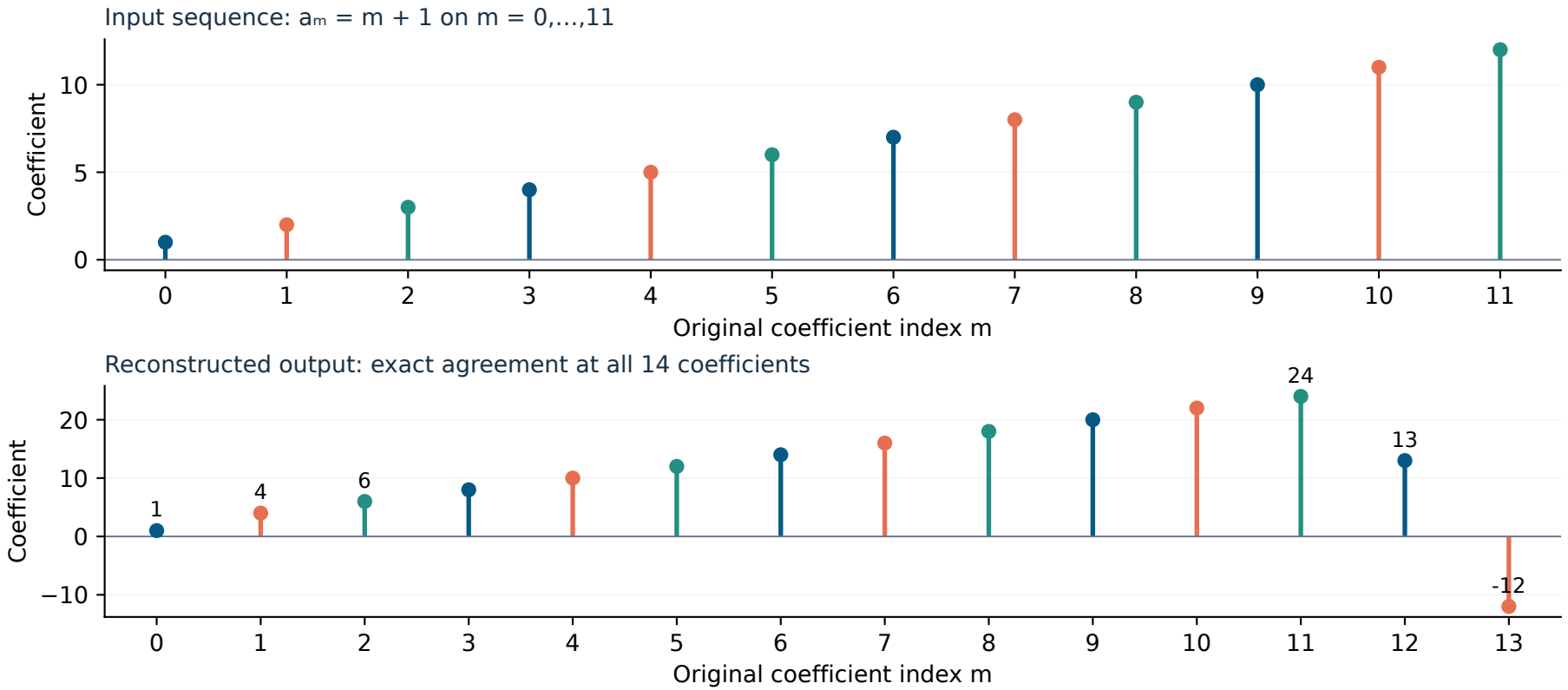
	X_0	X_1	X_2
Y_0	1	$-t$	$2t$
Y_1	2	1	$-t$
Y_2	-1	2	1

The t factors preserve the carry.

Input phase arrays

Phase 0: 1, 4, 7, 10

Phase 1: 2, 5, 8, 11

Phase 2: 3, 6, 9, 12

Filter the phases, then interleave.
Exact result equals direct convolution.